

# Annalisa Caligiuri

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📄 <https://scholar.google.com/citations?user=2l8ImdEAAAAJ&hl=en>

**Department:** Physics Department, University of the Balearic Islands (UIB)

**Position:** PostDoc

**Affiliation:** IFISC (Institute for Cross-Disciplinary Physics and Complex Systems) and UIB

## Education

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| Feb 2026 – present  | 📖 <b>Postdoctoral Researcher, IFISC and University of the Balearic Islands</b>                                                                                                                                                                   |
| Jan 2021 – Feb 2026 | 📖 <b>PhD Student, IFISC and University of the Balearic Islands</b> in Physics.<br>Supervisor: <i>Tobias Galla, Lucas Lacasa</i><br>Thesis title: <i>Complex systems with time-varying interactions.</i><br>Grade: <i>Cum laude</i>               |
| Apr 2024 – Jun 2024 | 📖 <b>Visiting Student, Queen Mary University of London.</b><br>Supervisor: <i>Vito Latora</i>                                                                                                                                                    |
| Sep 2017 – Jan 2020 | 📖 <b>M.Sc. in Physics, Sapienza University of Rome</b> (Theoretical Physics).<br>Thesis title: <i>Degree-ordered percolation on complex networks.</i><br>Supervisor: <i>Claudio Castellano</i><br>Grade: <i>110/110 with honors</i>              |
| Sep 2014 – Sep 2017 | 📖 <b>B.Sc. in Physics, Sapienza University of Rome.</b><br>Thesis title: <i>Analysis of chaotic systems with the Kolmogorov–Sinai entropy and Lyapunov exponents.</i><br>Supervisor: <i>Vittorio Loreto</i><br>Grade: <i>110/110 with honors</i> |

My research lies at the intersection of dynamical systems and network science. During my PhD, I have focused on the study of graph dynamical systems, where the object evolving in time is a network rather than a point in phase space. In particular, I investigate how concepts from dynamical systems theory—such as sensitivity to initial conditions and Lyapunov exponents—can be extended to temporal networks.

## Research Interests

Temporal networks, Chaos and nonlinear systems, Stochastic processes, Opinion dynamics Models

## Publication List

- 1 Caligiuri, A., Papo, D., Yener, G., Güntekin, B., Galla, T., Lacasa, L., & Zanin, M. (2026). Predictability of temporal network dynamics in normal ageing and brain pathology. *Journal of Complex Networks*, 14(1), cnafo58. 📄 doi:10.1093/comnet/cnaf058
- 2 Caligiuri, A., Galla, T., & Lacasa, L. (2025). Characterizing the dynamics of unlabeled temporal networks. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 35(5), 053122. 📄 doi:10.1063/5.0253870

- 3 Caligiuri, A., Emery, E., Ferreira, L., García-Castillo, J., Lindner, S. D., Molina-Hernández, J., ... Wang, B. et al. (2025). Time-varying ecological interactions characterise equilibrium and stability. *arXiv preprint arXiv:2506.22123*.
- 4 Caligiuri, A., & Galla, T. (2023). Noisy voter models in switching environments. *Phys. Rev. E*, 108, 044301. [doi:10.1103/PhysRevE.108.044301](https://doi.org/10.1103/PhysRevE.108.044301)
- 5 Caligiuri, A., Eguíluz, V. M., Di Gaetano, L., Galla, T., & Lacasa, L. (2023). Lyapunov exponents for temporal networks. *Phys. Rev. E*, 107, 044305. [doi:10.1103/PhysRevE.107.044305](https://doi.org/10.1103/PhysRevE.107.044305)
- 6 Caligiuri, A., & Castellano, C. (2020). Degree-ordered-percolation on uncorrelated networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2020(11), 113401. [doi:10.1088/1742-5468/abc1da](https://doi.org/10.1088/1742-5468/abc1da)

## Conferences

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- **NetSci2020** (First online conference): Poster with the title "Degree-Ordered-Percolation on complex networks";
- **FisEs Zaragoza, Spain** Poster with the title "Noisy Voter Model with time-varying influencers".
- **GAINs, Vienna** Attendance of summer school with the title 'Disintegration of societies' organized by the Santa Fe Institute in collaboration with Complexity Science Hub Vienna and other European institutions.
- **CCS22, Palma** Talk with the title "Noisy Voter Model with time-varying influencers".
- **NetSci23, Vienna** Talk with the title "Lyapunov Exponents for Temporal Networks".
- **NetPLACE@NetSci23, Vienna** organiser of NetPLACE@NetSci23
- **DD23, Naples** Talk with the title "Lyapunov Exponents for Temporal Networks".
- **Winter Workshop on Complex Systems 2024, Vall d'en Bas, Spain:** Worked on a small project using bees' proximity data.
- **1st NetPlace workshop on Networks & Physics (London, May 2025)** organiser.
- **NetSci25, Maastricht** Poster at TENET satellite.
- **Complexity72h 2025, Madrid** Preprint with the title: Time-varying ecological interactions characterise equilibrium and stability.
- **NetSciX26, Auckland** Talk with the title "Characterizing the dynamics of unlabeled temporal networks".
- **NetSciX26, Auckland** Moderator of the panel "Beyond Isolation: Getting Connected in Network Science" organized by NetPlace.

## Outreach

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- Ex organizer of NetPLACE, a community for the early-career researchers in Network Science.
- 11F International Day of Women and Girls in Science (in Spanish)

## Teaching

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- Physics I for the bachelor degree in Chemistry at University of the Balearic Islands (UIB): tutorial sessions (2023-2024).

## Skills

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| Languages           | 📖 | Italian (Mother Tongue), English (C1), Spanish (Fluent), French and German (Basic)                                                                 |
| Coding              | 📖 | C, Python, $\text{\LaTeX}$ , Matlab, Mathematica                                                                                                   |
| Theoretical methods | 📖 | Stochastic Processes, Path Integrals, Network Theory, Chaos Theory                                                                                 |
| Further knowledge   | 📖 | Online courses: <ul style="list-style-type: none"><li>• "Game theory"</li><li>• "Machine Learning with Python: A Practical Introduction"</li></ul> |
| Other interests     | 📖 | Outside my main research trajectory, I am a passionate indoor and outdoor climber and hold the certification of Indoor Climbing Instructor.        |

## References

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### **Dr Claudio Castellano**

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### **Dr Tobias Galla**

IFISC University of Balearic Islands,  
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### **Dr Lucas Lacasa**

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